PROJECT DOCUMENTATION

Website Traffic Analysis Dashboard

Website Analytics & User Behaviour Insights · Built with Microsoft Power BI

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Topic	Website Traffic Analysis
Project	Website Traffic Analysis Dashboard
Tool	Microsoft Power BI & SQL

GITHUB REPOSITORY

https://github.com/ankitpandey5776/network-traffic-analyzer

1. Problem Statement

Modern websites generate large volumes of session, traffic, and engagement data on a daily basis. This raw data, stored across flat files and relational databases, offers little business value to stakeholders unless it is transformed into actionable visual insight. Website managers and digital marketers need clear answers to practical questions: which devices drive the most engagement, which traffic sources produce quality sessions, where bounce rates are highest, and how page-view patterns shift across quarters.

Answering these questions manually from raw CSV or SQL tables is time-consuming and error-prone. The goal of this project is to build an interactive, multi-page Power BI dashboard on top of a structured website traffic dataset. The dashboard enables any stakeholder, without writing a single query, to filter data by device, traffic source, browser, region, and content segment, and instantly see session trends, bounce rates, page views, and geographical performance at a glance.

2. Solution and Features

The project delivers a three-page interactive dashboard. Page 1 covers overall session and bounce rate trends with a traffic-source heat map. Page 2 shows geographical city-wise and region-wise breakdowns. Page 3 provides monthly and daily trend analysis with device-level sparklines. All pages share slicer filters (Device, Source) so selections flow across the full report.

Key Features

- **Five KPI Cards:** AVG Page Views, AVG Session Duration, Total Page Views, Total Session Duration, Total Sessions
- **Interactive Slicers:** Device type (Desktop, Laptop, Mobile, Tablet, Other) and Traffic Source (Email, Organic, Paid, Referral, Social)
- **Session Duration Analysis:** Quarterly trends for total and average session duration across 2020–2021
- **Bounce Rate Trends:** Overall quarterly bounce rate with device, browser, content-segment, and traffic-source breakdowns
- **Page Views Analysis:** Total page views quarterly trend plus city-wise and region-wise comparison bar charts
- **Geographical Insights:** Top-5 cities ranked by session duration, page views, bounce rate, and total sessions; region donut charts
- **Traffic Source Heat Map:** Colour-coded matrix comparing Total Sessions, Session Duration, Avg Duration, Page Views across all sources
- **Monthly & Daily Trends:** Line charts for bounce rate, avg session duration, and page views by month and by day of week
- **Device Sparklines:** Small multiples showing quarterly session trends per device type and per traffic source

3. Key Metrics Tracked

5,200+ Total Sessions	**0.19 hrs** Avg Session Duration	**2,522** Total Page Views	**9.58%** Bounce Rate
6 Types Traffic Sources	**5 Types** Devices Tracked	**4 Zones** Regions Analyzed	**4 Types** Browsers Tracked

Metric Definitions

- **Session Duration:** Total and average time (in hours) users spend on the website per session
- **Bounce Rate:** Percentage of single-page sessions where users left without further interaction
- **Total Page Views:** Aggregate number of pages viewed across all sessions in the analysis period
- **Traffic by Device:** Session statistics split by Desktop, Laptop, Mobile, Tablet, and Other
- **Traffic Sources:** Page views, session duration, and bounce rate by Organic, Email, Social, Paid, Referral, Other
- **Geographical Analysis:** Session and engagement metrics broken down by Indian region and top cities

4. Dashboard Screenshots

The screenshots below show the final three-page dashboard built in Microsoft Power BI Desktop. All pages share a consistent dark navy visual theme with unified slicer panels, cross-page filter synchronisation, and KPI cards for a clean executive finish.

Page 1 — Overall Traffic & Trends

Figure 1: Overall Traffic — session duration, bounce rate, page views, and traffic-source heat map.

Page 2 — Geographical Insights

Figure 2: Geographical Insights — top-5 cities and region-level donut charts.

Page 3 — Monthly & Daily Trend Analysis

Figure 3: Monthly & Daily Trends — bounce rate, session duration, page views, and device sparklines.

5. Dashboard Pages

Page 1 — Overall Traffic & Trends

- Total Session Duration and AVG Session Duration quarterly area charts (2020 Q1 – 2021 Q4)
- Bounce Rate quarterly trend line chart highlighting peaks and recovery
- Total Page Views quarterly trend with data labels
- Traffic-source heat map (darkest blue = highest, lightest cream = lowest)
- Bounce Rate by Device bar chart, by Browser bar chart, and by Content Segment card list
- Gauge visual showing overall Bounce Rate at 9.58%

Page 2 — Geographical Insights

- Top-5 cities by Total Session Duration — Chennai leads with 20 hrs
- Top-5 cities by Total Page Views — Chennai tops with 371 views
- Top-5 cities by Bounce Rate — Chennai 16.22%, Ahmedabad 15.15%, Pune 14.58%
- Top-5 cities by Total Sessions — Chennai 111, Hyderabad 89, Kolkata 87
- Region-level donut charts for Session Duration, Bounce Rate, and Page Views (South 34.55% leads)

Page 3 — Monthly & Daily Trend Analysis

- Bounce Rate monthly trend: peaked at 18.33% in March, lowest at 4.84% in July and Oct–Dec
- Bounce Rate daily trend: highest on Sunday (17.31%), lowest on Tuesday (4.81%)
- Average Session Duration monthly and daily trend line charts
- Total Page Views monthly trend — highest in August (235), lowest in February (182)
- Total Sessions by Device Type — quarterly sparkline grid for all 5 device types
- Total Sessions by Traffic Source — quarterly sparkline grid for all 6 traffic sources

6. Technology Stack

Technology	Purpose	Details
Microsoft Power BI	Dashboard & Visualization	Primary BI Tool
SQL (MySQL/PostgreSQL)	Data Extraction	Complex Queries & Aggregations
Power Query (M)	Data Transformation	Cleaning, Shaping, Joins
DAX	Calculated Measures	KPIs, Bounce Rate, Averages
CSV Flat Files	Data Source	Traffic, Device, Geo Lookups

Data Preparation

The dataset comprised multiple CSV files — Website_Traffic_Data.csv (fact table), Device_Lookup.csv, Geo_Lookup.csv, and Source_Lookup.csv (dimension tables). SQL queries were written to extract, aggregate, and join tables before loading into Power BI. Power Query was used for type conversion, relationship setup, and creation of a dedicated Date Table. DAX measures were authored for all KPI cards, trend lines, and conditional formatting thresholds.

7. Unique Points

- **Three-page narrative flow:** Page 1 answers overall trends, Page 2 answers where users are, Page 3 answers when users engage — matching how stakeholders read traffic reports
- **SQL-driven data layer:** All aggregations were first prototyped in SQL queries before being replicated as DAX measures, ensuring correctness and transparency
- **Colour-encoded heat map:** The traffic-source matrix uses a custom blue-to-cream diverging scale, making high and low performers instantly identifiable without reading numbers
- **Device and source sparklines:** Small-multiple area charts on Page 3 allow simultaneous comparison of all device types and traffic sources without cluttering a single chart
- **Geographical depth:** Four separate bar charts for cities plus three donut charts for regions provide both micro (city) and macro (region) geographical perspectives
- **Cross-page filter sync:** Device and Source slicers in the left panel update all three pages simultaneously; users never lose filter context while navigating
- **Consistent dark theme:** The dark navy background with high-contrast amber/red titles and blue charts ensures the dashboard is readable in both boardroom presentations and screen shares

8. Key Insights from the Data

- **Desktop dominates:** Desktop users account for the highest bounce rate at 11.54%, suggesting potential UX improvements needed for desktop layouts
- **Bounce rate improving:** The quarterly bounce rate trend shows a sharp decline from 13.33% in Q1 2021 to just 3.26% by Q4 2021, indicating significantly improving content engagement
- **Chennai leads geographically:** Chennai ranks first across session duration (20 hrs), page views (371), and total sessions (111), making it the most engaged city
- **Referral drives highest bounce:** Referral traffic shows the highest bounce rate (12.28%) while Organic shows the lowest (5.43%), confirming that SEO-driven traffic produces the most engaged visitors
- **Page views growing:** Total page views grew steadily from 308 in Q1 2020 to 335 in Q4 2021, reflecting consistent audience growth
- **South region leads:** The South region accounts for 34.55% of total session duration — the highest share among all four regions — making it the primary market
- **Friday peak engagement:** Daily trend analysis shows Friday as the peak day for page views (381) and Thursday for bounce rate (12.38%), suggesting weekday content strategy opportunities

9. Future Improvements

- **Live data connection:** Replace CSV imports with scheduled refresh from a cloud SQL database (Azure SQL or BigQuery) for near-real-time analytics
- **Predictive analytics:** Integrate Power BI's built-in forecasting or Python/R scripts to project session growth, bounce rate trajectory, and seasonal patterns

- **User-level analysis:** Incorporate user-ID data to build cohort analysis, return visitor rates, and customer journey maps
- **Drill-through pages:** Add hidden drill-through pages for each city and device type, enabling one-click deep dives from any summary chart
- **Row-Level Security (RLS):** Implement role-based access so regional managers see only their own geography's data
- **Publish to Power BI Service:** Deploy to Power BI Service with scheduled refresh and automated email subscriptions for weekly traffic summaries
- **Mobile-optimised layout:** Create a dedicated phone-view layout so the dashboard remains readable on mobile devices for on-the-go stakeholders
- **A/B testing integration:** Connect to Google Optimize or a similar platform to correlate A/B test results with bounce rate and session duration changes

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